

Functional Annotation Report: *mrdA-II* (PBP-2) in *Pseudomonas putida* KT2440

Gene: `mrdA-II` (synonym `mrdA`; ordered locus **PP_4807**) **Protein:** Peptidoglycan D,D-transpeptidase MrdA / Penicillin-binding protein 2 (PBP-2) **UniProt:** Q88DL8 · **EC:** 3.4.16.4 **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125), a Gram-negative, rod-shaped soil γ -proteobacterium **Family:** Transpeptidase family, MrdA subfamily (Class B high-molecular-mass PBP) **Signature domains:** Penicillin-binding, transpeptidase (IPR001460); Beta-lactam/transpeptidase-like (IPR012338, IPR050515); PBP dimerization "pedestal" domain (IPR005311, IPR036138)

1. Identity verification

The gene symbol, the UniProt description, the assigned EC number (3.4.16.4, serine-type D-Ala-D-Ala transpeptidase/carboxypeptidase activity), and the InterPro domain complement are **mutually consistent** and unambiguously identify Q88DL8 as a **Class B penicillin-binding protein 2 (PBP2/MrdA)**. The PCN-bd_Tpept (IPR001460) transpeptidase domain plus the PBP_dimer "pedestal" domain (IPR005311/IPR036138) is the diagnostic two-module architecture of the MrdA/PBP2 subfamily. No conflicting gene-symbol ambiguity was encountered.

One organism-specific point, established here directly: the symbol is `mrdA-**II**` because *P. putida* KT2440 encodes **two** paralogous PBP2/MrdA transpeptidases. A UniProt proteome query (taxid 160488) returns **mrdA-I = PP_3741** (Q88GI2, 631 aa) and **mrdA-II = PP_4807** (Q88DL8, 629 aa; the target), alongside the divisome transpeptidase **FtsI/PBP3** (PP_1331) and the bifunctional class A PBPs **PBP1A** (*mrcA*/PP_5084) and **PBP1B** (*mrcB*/PP_4683). A BLOSUM62 global alignment shows the two PBP2 paralogs are **70.8% identical** (464/655 aligned positions) — high similarity but clearly distinct, indicating a retained, functional gene duplication rather than a recent identical copy or a pseudogene.

Importantly, despite the "-II" suffix, the target PP_4807 is the *genomically canonical elongation PBP2*. Its gene sits in the classic *mrd* operon: the immediately upstream gene PP_4806 = *mrdB/rodA* (Q88DL9) encodes the SEDS peptidoglycan glycosyltransferase RodA

(EC 2.4.99.28) — the obligate partner of PBP2 in the elongasome — with **PP_4805 = *mltB*** (a lytic transglycosylase) just beyond it. This *mrdA*–*mrdB*(*rodA*) tandem mirrors the *E. coli* *mrdAB* operon. The paralog *mrdA-I* (PP_3741), by contrast, is **not** in an *mrd* operon (its neighbours are a major-facilitator transporter, a porin-like protein and a glutathione S-transferase, with no adjacent *rodA*), marking it as the divergent, orphan/accessory copy. Thus ***mrdA-II* is the PBP2 physically coupled to its cognate RodA**, and the functional description below applies to it with organism-specific genomic support. Direct wet-lab enzymology of PP_4807 in isolation is not yet published, so cross-organism family biochemistry/structure supplies mechanistic detail; the diagnostic catalytic features are nonetheless confirmed **directly on the Q88DL8 sequence** (§2).

2. Primary molecular function — the catalyzed reaction and substrate specificity

MrdA/PBP2 is a **serine D,D-transpeptidase** that catalyzes the **crosslinking step of peptidoglycan (PG, murein) synthesis**. "Penicillin-binding protein 2 (PBP2) plays a key role in the formation of peptidoglycans in bacterial cell walls by crosslinking glycan chains through transpeptidase activity" (Choi et al. 2024,).

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- **Reaction:** a D,D-transpeptidation. The enzyme's active-site serine attacks the terminal **D-Ala–D-Ala** peptide bond of a donor stem peptide, forming an acyl-enzyme intermediate and releasing the terminal D-Ala. The acyl group is then transferred to the free amino group of the **meso-diaminopimelate (mDAP)** residue at position 3 of an acceptor stem peptide on a neighbouring glycan strand, producing a **4→3 (D-Ala–mDAP) crosslink**. In Gram-negative γ -proteobacteria such as *P. putida* the acceptor is mDAP, giving the canonical A1 γ (DAP-type) murein.
- **Substrate:** the peptide side-chains of nascent **Lipid II–derived glycan strands** (GlcNAc–MurNAc backbone bearing the pentapeptide L-Ala–D-Glu–mDAP–D-Ala–D-Ala). The natural substrate is therefore the polymerized, membrane-associated glycan, not free Lipid II.
- **Structure–function basis:** crystal structures of family members show "an elongated two-domain structure, consisting of a transpeptidase domain and a pedestal domain, and ... typical active site residues necessary for transpeptidase activity, as observed in other PBP2 proteins" (Choi et al. 2024,

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The transpeptidase domain carries the conserved SxxK, S/YxN and K(T/S)G catalytic motifs; the SxxK serine is the nucleophile.

- **Confirmed directly on Q88DL8 (this study, sequence analysis):** the 629-aa target protein contains, in the correct order and spacing, all diagnostic elements: an N-terminal cytoplasmic tail (res 1–20) → single transmembrane helix (res 21–41) → pedestal/dimerization domain (res 63–236, Pfam PF03717) → transpeptidase domain (res 268–604, Pfam PF00905), with the **SxxK nucleophile motif S327-T-I-K330** (UniProt annotates Ser327 as the acyl-ester intermediate) and the **K[S/T]G motif G537-K-S-G540** that lines the substrate/ β -lactam pocket. InterPro assigns the subfamily-specific signature **IPR017790 "Penicillin-binding protein 2."** These features are present in the actual sequence, so Ser327 is the predicted catalytic nucleophile of *this* enzyme.
- **Pharmacological signature (mechanistic evidence of function):** because the catalytic serine mimics the D-Ala–D-Ala substrate, **β -lactam antibiotics acylate and irreversibly inhibit it** — this is the "penicillin-binding" property. PBP2 is the specific target of the amidinopenicillin **mecillinam (amdinocillin)**: "amdinocillin (mecillinam) show[s] selectivity for PBP2" (Kocaoglu & Carlson 2015,

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and "the beta-lactam mecillinam specifically targets the PBP2 enzyme in the cell elongation machinery of *Escherichia coli*" (Lai et al. 2017,

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In pseudomonads specifically, PBP2 is the antibacterial target of the diazabicyclooctane " β -lactam enhancers" zidebactam/WCK 5153 (Moya et al. 2017,

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and a PBP2 point mutation (V516M) confers zidebactam resistance in *Pseudomonas* (Sastre-Femenia et al. 2026,

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— direct evidence that PBP2 is the physiological, drug-relevant transpeptidase in this genus.

3. Biological process — cell elongation via the Rod complex (elongasome)

MrdA/PBP2 is the **cognate transpeptidase of the cell-elongation ("Rod") peptidoglycan synthase**, working in an obligate partnership with the SEDS-family glycosyltransferase **RodA**:

- **RodA–PBP2 is a two-enzyme PG synthase.** A SEDS glycosyltransferase and a Class B PBP "form the core of the multi-protein complex required for PG assembly" (Nygaard et al. 2023,

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cryo-EM of the *E. coli* RodA–PBP2 complex). RodA polymerizes Lipid II into a nascent glycan strand; the strand is then delivered to the PBP2 transpeptidase site for crosslinking into the existing sacculus, "functionally linking these two central enzymatic activities required for cell wall peptidoglycan biosynthesis"

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Thus **glycan polymerization (RodA) and peptide crosslinking (PBP2/MrdA) are physically coupled.**

- **The synthase is organized by the MreB cytoskeleton.** The elongasome/Rod complex "consists of six highly conserved proteins, including the actin-like MreB, the PG synthase complex RodA-PBP2, and three regulatory membrane proteins, MreC, MreD, and RodZ" (Zhan et al. 2026,

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MreB "is localized beneath the cytoplasmic membrane, where it organizes the elongasome complex" (Maharjan et al. 2026,

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PBP2 is coupled directly to this cytoskeleton: "both the cytoplasmic region of PBP2 and the C-terminal tail of RodA interact with MreB ... disruption of these interactions results in a loss of rod shape" (Zhan et al. 2026,

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- **Regulation.** The activity of the RodA–PBP2 synthase is controlled by MreC/MreD and RodZ; "RodZ acts through MreBCD to activate the elongasome" (Zhan et al. 2026,

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In *B. subtilis* the processive circumferential motion of the complex is tuned by RodA levels via a "molecular motor tug-of-war" (Middlemiss et al. 2024,).

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- **Physiological role and redundancy.** New PG is inserted into the **lateral (side) wall** to extend the cylinder of the rod; PBP2 crosslinking is what converts the newly polymerized glycan into load-bearing wall. Inhibition or loss of PBP2 abolishes rod shape (cells round up) and, under growth, leads to lysis. This elongation function is partly buffered by the class A PBP1B: PBP2 inhibition "can be compensated for by the presence of the class A penicillin binding protein, PBP1B" (Grinnell et al. 2022,),

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and endopeptidases that cleave existing crosslinks stimulate new synthesis by the Rod system (Lai et al. 2017,).

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This division of labour distinguishes MrdA/PBP2 (elongation) from PBP3/FtsI (septation/division), which is the cognate transpeptidase of the FtsW–PBP3 divisome.

- **Organism-specific genomic evidence that *mrda-II* is the RodA-coupled elongation PBP2.** In *P. putida* KT2440, the target gene *mrda-II* (PP_4807) is the second gene of an *mrda* operon whose first gene, **PP_4806 = *mrdaB/rodA*** (Q88DL9), encodes the **SEDS peptidoglycan glycosyltransferase RodA** (EC 2.4.99.28) — precisely the obligate synthase partner that, in the cryo-EM structure (

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polymerizes the glycan and delivers it to the PBP2 transpeptidase site. This tight genomic linkage of *mrda-II* to its cognate *rodA* (mirroring the *E. coli* *mrdaB* operon) is direct, organism-specific support that PP_4807 is the elongation-specific PBP2 that assembles the **RodA–PBP2 synthase** in this bacterium. The other PBP2 paralog, *mrda-I* (PP_3741), lacks an adjacent *rodA* and is the divergent/accessory copy.

- ***P. putida* context.** *P. putida* is a rod-shaped organism whose DAP-type peptidoglycan is subject to chemical editing by environmental D-amino acids; work in this species showed that non-canonical D-amino acids (e.g. D-canavanine) can be incorporated and perturb crosslinking and division, with resistance mapping to a division transpeptidase (Aliashkevich et al. 2021,).

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This confirms that *P. putida* uses the canonical PBP transpeptidase machinery for wall crosslinking, the elongation arm of which is MrdA/PBP2.

4. Subcellular localization

MrdA/PBP2 is a **bitopic (single-pass) inner-membrane (cytoplasmic-membrane) protein**:

- **Direct annotation:** UniProt (HAMAP-Rule MF_02081) assigns Q88DL8 to the "**Cell inner membrane; Single-pass membrane protein.**" UniProt sequence features locate the single transmembrane helix at **residues 21–41**.
- **Topology (confirmed on Q88DL8 + family-conserved):** a short N-terminal **cytoplasmic tail (res 1–20)**, a single **non-cleaved transmembrane anchor (res 21–41)**, and a large **periplasmic region (res ~42–629)** comprising the pedestal (dimerization) domain (63–236) and the C-terminal transpeptidase domain (268–604).
- **Where catalysis happens:** Lipid II is flipped to the periplasmic (outer) face of the inner membrane, so **the transpeptidase reaction occurs in the periplasm**, on the outer surface of the cytoplasmic membrane, immediately adjacent to the growing sacculus.
- **Where it is positioned:** the enzyme is recruited to discrete, circumferentially moving foci along the lateral cell body by the MreB filament, which lies "beneath the cytoplasmic membrane"

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The cytoplasmic N-terminus of PBP2 contacts MreB

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so the single membrane pass allows one molecule to **couple cytoplasmic cytoskeletal positioning to periplasmic wall synthesis**.

Net localization statement: **cytoplasmic (inner) membrane, single-pass, with the catalytic domain projecting into the periplasm; concentrated in mobile lateral-wall elongasome complexes.**

5. Pathway summary

Peptidoglycan biosynthesis, elongation branch:

UDP-MurNAc-pentapeptide → **Lipid I** → **Lipid II** (cytoplasmic/inner-leaflet steps, MraY/MurG) → flip to periplasmic face (MurJ) → **RodA glycosyltransfer** (glycan-strand polymerization) → **MrdA/PBP2 D,D-transpeptidation** (4→3 mDAP–D-Ala crosslinking) → mature, load-bearing sacculus. Spatial control: **MreB–MreC–MreD–RodZ** (Rod complex) direct where and when RodA–PBP2 acts.

MrdA/PBP2 sits at the **crosslinking node** of this pathway, downstream of glycan polymerization and immediately determining the mechanical integrity and rod geometry of the wall.

6. Evidence quality and hierarchy

- **Direct enzymatic/structural evidence (family, high confidence):** crystal structure and transpeptidase active-site definition of PBP2

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 cryo-EM mechanism of the RodA–PBP2 synthase

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- **Direct genetic/physiological evidence (family, high confidence):** mecillinam/zidebactam target specificity for PBP2

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 MreB–PBP2 coupling and elongasome organization

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redundancy with PBP1B

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- **Direct sequence/bioinformatic evidence on Q88DL8 (this study):** the target sequence carries the intact SxxK (Ser327) and K[S/T]G (537–540) catalytic motifs, the single TM anchor (21–41), and the pedestal + transpeptidase domain pair (Pfam PF03717/PF00905; InterPro IPR017790 "PBP2"). Global alignment establishes *mrda*-II and *mrda*-I (PP_3741) as 70.8%-identical paralogs, both catalytically intact — confirming PP_4807 is a functional, not degenerate, PBP2.

- **Direct genomic-context evidence on PP_4807 (this study):** *mrda*-II is the second gene of an *mrda* operon with its cognate SEDS partner *mrdaB/rodA* (PP_4806) immediately upstream, marking it as the RodA-coupled, elongation-specific PBP2 in *P. putida* (whereas *mrda*-I/PP_3741 has no adjacent *rodA*).

- ***P. putida*-specific supporting evidence:** DAP-type PG editing and transpeptidase involvement in this organism

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UniProt "Cell inner membrane; Single-pass membrane protein" annotation for Q88DL8 (HAMAP MF_02081).

- **Inference (annotation-level):** the *physiological* details specific to PP_4807 in cells (essentiality, kinetics, whether it is the principal vs. accessory elongation PBP2, condition-specific deployment) are **not yet measured directly on this locus** and are inferred from family conservation.

7. Limitations and future directions

1. **Paralog specificity.** *P. putida* KT2440 encodes two *mrda*/PBP2 genes — *mrda*-I (PP_3741) and *mrda*-II (PP_4807, this target) — that are 70.8% identical and both catalytically intact. Genomic context argues that ***mrda*-II is the canonical elongation PBP2** (operon-linked to its RodA partner) and *mrda*-I the accessory copy, but this has not been tested functionally: the paralogs' relative essentiality, expression conditions, and differential β -lactam/mecillinam sensitivity remain unknown. Targeted single- and double-deletion/

complementation of PP_4807 vs PP_3741, Bocillin-FL PBP profiling, and mecillinam-sensitivity assays would resolve this.

2. **Direct biochemistry** (kinetics, Lipid II crosslinking assays, β -lactam acylation rates) on the *P. putida* enzyme itself is lacking.
3. **In-cell localization/dynamics** (fluorescent-fusion tracking of PP_4807 relative to MreB and the second paralog) would confirm elongasome membership for this specific gene product.

8. Conclusion

mrda-II (PP_4807, Q88DL8) encodes a **Class B penicillin-binding protein 2 (PBP2/MrdA)**, a **single-pass inner-membrane serine D,D-transpeptidase (EC 3.4.16.4; catalytic nucleophile Ser327)** whose primary function is to **crosslink peptidoglycan (4→3 D-Ala-meso-DAP bonds) during cell-wall elongation**. It acts in the **periplasm** as the **cognate transpeptidase of the RodA-PBP2 synthase**, which is polymerized-glycan-fed by the SEDS enzyme RodA and spatially organized by the **MreB/MreC/MreD/RodZ elongasome**, thereby inserting new lateral wall material to maintain the rod shape of *P. putida*. It is the physiological target of the PBP2-specific agents mecillinam and zidebactam. Notably, it is **one of two PBP2 paralogs** in KT2440 (with *mrda-I*/PP_3741, 70.8% identical); both are catalytically intact. Genomic context marks ***mrda-II* (PP_4807) as the canonical elongation PBP2**, since it is encoded in an *mrda* operon directly downstream of its cognate SEDS glycosyltransferase partner *rodA/mrdB* (PP_4806), whereas *mrda-I* lacks an adjacent *rodA*. The precise in-cell division of labour between the two paralogs remains to be tested experimentally.